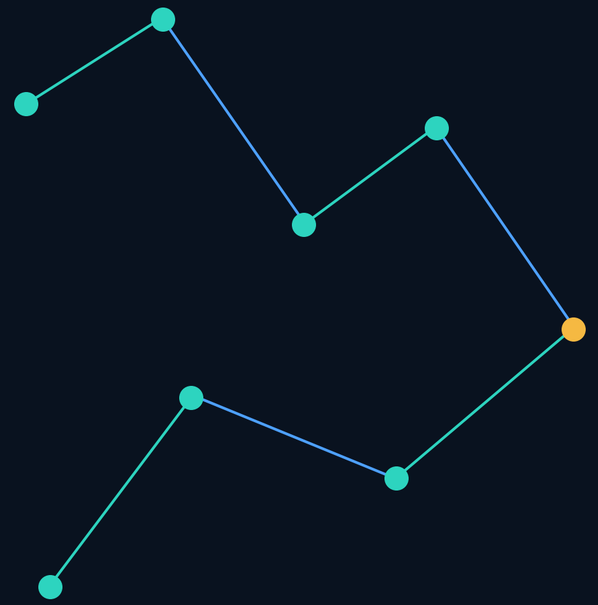
EWMA Covariate Shift Demo



Research progress review

Auditable CSE Reconstruction and Validation for Motor-Imagery EEG

Classifier Benchmarks, Complementary PCA Monitoring and Reproducibility Analysis

Haocheng Lin

20th October 2026

Ulster University

Research narrative and evidence boundaries

01	**Problem and literature**	EEG non-stationarity; research lineage; clinical evidence: BCI Competition Datasets
02	**Representation and detection**	FBCSP features; frozen PCA coordinates; EWMA and Stage II
03	**Adaptation and system policies**	Updated SVM, label availability, Diethe, and passive baselines
04	**Validation and research plan**	BCI datasets, reproducibility gaps, pilot evidence, and next milestones

From published algorithms to an auditable application

Data preparation → shift evidence → adaptation policy → reproducible results

CORE COMPUTATIONAL PATH

CANDIDATE ≠ CONFIRMED ≠ ADAPTATION

01

DATA SOURCES

BCI IV 2A / 2B
Synthetic + turbofan

src/bci/ · src/simulation/

02

REPRESENTATION

Trial extraction · FBCSP
Train-fitted PCA

src/bci/fbcsp.py

03

STAGE-I SCREEN

SD / MSD–EWMA
PCA score + residual

src/detection/stage1/

04

STAGE-II TEST

K–S / Hotelling variants
Opt-in MMD + GPR

src/detection/stage2/

05

ADAPTATION

Decision policies
SVM · PWKNN · bagging

src/adaptation/

06

EVIDENCE + UI

Metrics / CSV / Markdown
Flask · GitHub Pages

src/reporting/ · src/web/

WHAT IS ALREADY ACHIEVED

Concrete deliverables backed by source modules and saved evidence

01 **REPRODUCTION FOUNDATION**

Raza 2015 synthetic D2/D3/D4 studies
BCI IV 2A/2B Table 1 processing and comparison outputs.

src/experiments/paper2015/ · scripts/run_bci_table1_reproduction.py

02 **DIAGNOSTIC BREADTH**

Stage-I/II evidence kept separate; complementary PCA monitoring, Park CUSUM, and opt-in GPR/MMD.

src/detection/ · docs/gpr_kernel_validation.md

03 **WORKING RESEARCH PLATFORM**

Supervised and transductive experiments; Diethe policy laboratory; Flask results hub, tests and CI.

src/adaptation/ · src/web/ · tests/

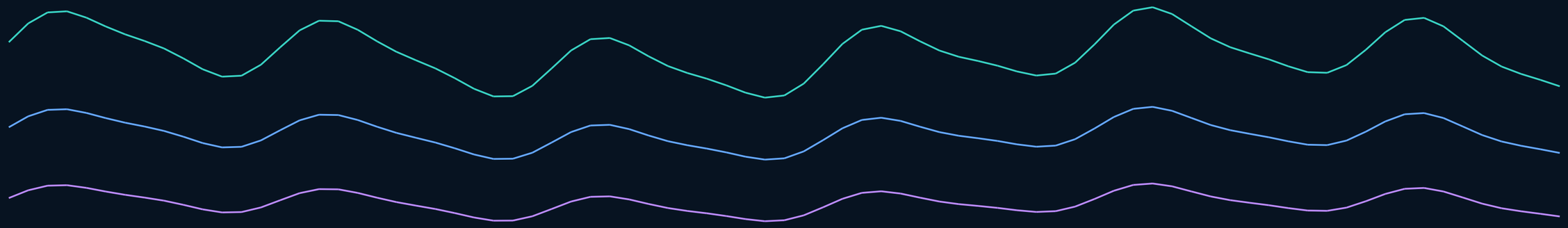
RESEARCH BOUNDARY • Fully integrated digital twin, delayed-label joining and dynamic CSE-UAEL ensemble remain future work.

Source: github.com/hlin863/ewma_covariate_shift_demo

LITERATURE REVIEW · OCTOBER 2026

From detecting shift to self-maintaining learning

A source-grounded map of methods, code and open research questions



01 REPRESENT
CSP · FBCSP

02 MONITOR
EWMA · CUSUM · ABCD

03 DECIDE
Validation · selective update

04 RETAIN
Continual learning

Based on the WP1 literature library, the 19 Oct diary audit, and hlin863/ewma_covariate_shift_demo

RESEARCH MAP

Six functional layers, one system question

01 REPRESENT

Motor imagery EEG / CSP / tangent space

Ramoser 2000 · Gaur 2019

02 DETECT

Online EWMA warning, CUSUM, subspace change

Raza 2015 · Park 2023 · Heyden 2024

03 VALIDATE

KS / Hotelling T^2; false alarm vs delay

Raza 2015 · Alippi 2017

04 ADAPT

CSD-triggered updates and ensembles

Raza 2016 · Chowdhury 2018 · Raza 2019

05 RETAIN

Continual learning and stability–plasticity

Wang 2022 · Kudithipudi 2022

06 OPERATE

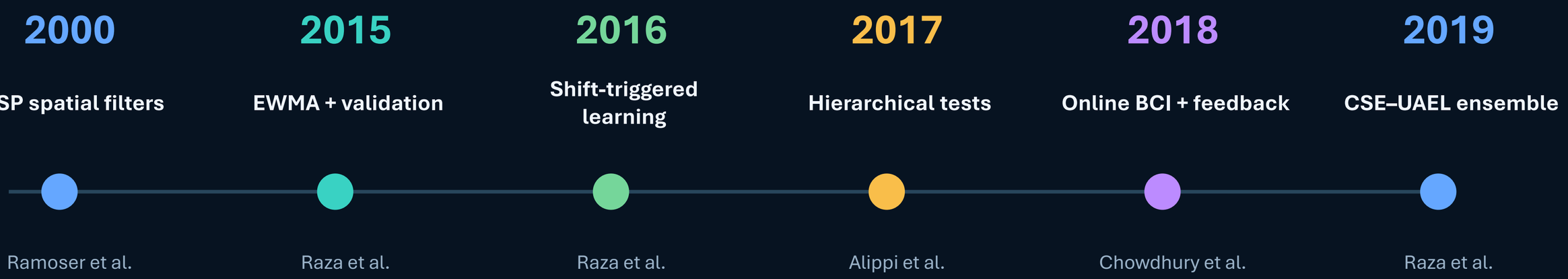
Policy engine, PdM, digital twins

Diethe 2019 · Hurtado 2023 · Gupta 2024

CANONICAL LINEAGE

The papers that shaped the BCI detector

A selective source-to-method lineage; later literature extends rather than supersedes it.



PhD reconstruction point trial-aligned EEG → FBCSP/PCA → warning → validation → adaptation policy

DIRECT PREDECESSORS

Four cornerstone papers & what each contributes

Prioritise the paper's functional role, the reusable method, and its remaining evidential boundary.

Ramoser et al. (2000)

EEG SIGNAL REPRESENTATION

Common spatial patterns for discriminatory single-trial hand imagery.

REPO: Maps to upstream CSP/FBCSP features.

LIMIT: CSP accuracy is not drift-detection accuracy.

Raza et al. (2015)

SEQUENTIAL CHANGE TEST

Stage I EWMA; Stage II K–S (univariate) or Hotelling T^2 (multivariate).

REPO: Maps to CSE warning and validation.

LIMIT: Two-stage alarms require calibration and chronology audit.

Chowdhury et al. (2018)

ONLINE CLOSED-LOOP APPLICATION

CSD-based adaptation linked to visual / hand-exoskeleton feedback.

REPO: Maps to BCI translational architecture.

LIMIT: Clinical findings do not validate this repo's online deployment.

Raza et al. (2019)

ADAPTIVE ENSEMBLE FRAMEWORK

CSE-UAEL: conditional ensemble expansion after estimated shifts.

REPO: Maps to Table 1 reconstruction / adaptation pathway.

LIMIT: Published EEG counts remain unreconciled.

NEW STATISTICAL DIRECTIONS

Different tests answer different change questions

A useful comparison is statistical target, not simply whether a method raises an alarm.

Raza et al. (2015)

TWO-STAGE EWMA / T^2

SIGNALS	CSP/PCA feature stream
STAGE I	Online EWMA warning
STAGE II	K–S or two-window Hotelling T^2
WHAT IT SUPPORTS	Mean-sensitive multivariate validation
OPEN IN PHD	Repeated testing, covariance assumptions

Park et al. (2023)

STRUCTURAL CHANGE

FIRST-ORDER	Mean CUSUM / peak agreement
SECOND-ORDER	Wavelet spectral matrices / dynamic PCA
DISTINCTION	Covariance + temporal dependence changes
IN REPO	Mean-CUSUM and proposed CPSS
NOT YET	Second-order method / paper reproduction

Heyden et al. (2024)

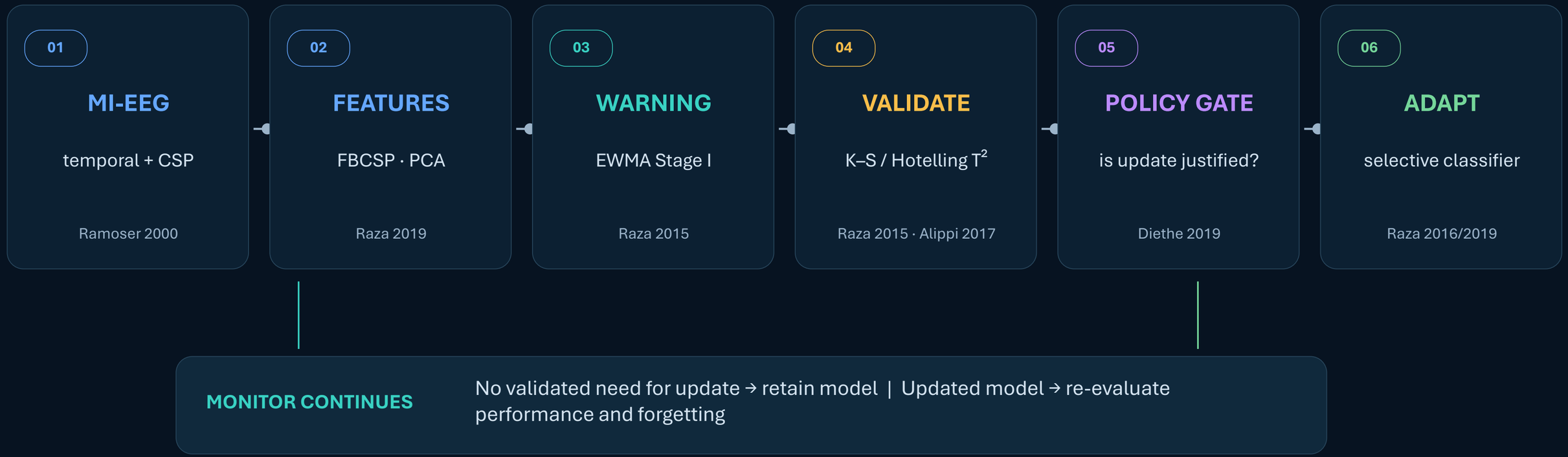
HIGH-DIMENSIONAL STREAMS

DETECTOR	ABCD / adaptive window
SIGNAL	Encoder–decoder reconstruction loss
TARGETS	Change time, subspace, severity
VALUE	Complementary comparison candidate
NOT ESTABLISHED	ABCD EEG benchmark in this repo

Caution: Hotelling mean-vector evidence does not establish general second-order or concept-change detection.

Where the literature enters the algorithm

A proposed integration schematic—not a claim that every downstream component is deployed.



Online exoskeleton feedback: demonstrated by Chowdhury et al. (2018).

BEYOND CSE

Why drift detection alone is not the PhD end-point

Separate monitoring evidence, adaptation policy, knowledge retention and deployment outcomes.

MONITOR

Diethe et al. (2019)

Data + prediction monitoring; model-policy engine; validate and roll back.

Repo: six-policy laboratory

RETAIN

Wang et al. (2022)

L2P / DualPrompt use lightweight prompts to address rehearsal-free CL.

Conceptual comparator; not EEG result

EVALUATE

Hurtado et al. (2023)

Realistic PdM CL benchmarks and the cost of static-distribution assumptions.

Cross-domain research motivation

DEPLOY

Gupta et al. (2024)

TWIN-ADAPT: continual anomaly-classifier updates in smart-lab digital twins.

Industrial application direction

PROPOSED NOVELTY TEST: Does an evidence- and cost-aware gate beat established update baselines at matched alarm budgets?

EVIDENCE LEDGER

A visible boundary between work done and work proposed

Evidence status from the 19 October Notion audit and latest connected GitHub commit history.

IMPLEMENTED / AUDITABLE

- **EEG representation**
  Trial-aligned 2A/2B, FBCSP / PCA
- **CSE baseline**
  EWMA warnings + Hotelling variants
- **Monitoring extensions**
  PCA score/residual; Park mean CUSUM
- **Research infrastructure**
  CLI, tests, saved outputs, pages

EXPLORATORY EVIDENCE

- **CPSS paired pilot**
  24 seeds per stationary/shift case
- **Weak mean shift**
  6/24 Park vs 11/24 CPSS alarms
- **False alarms**
  0/24 Park vs 1/24 CPSS stationary
- **Not inferential**
  Null calibration threshold = 0

NOT YET ESTABLISHED

- **Published Raza counts**
  2A/2B discrepancies unreconciled
- **Park second order**
  Wavelet spectral / dynamic PCA future
- **Full CSE–UAEL**
  Original adaptive ensemble not reproduced
- **CL/industrial claims**
  No demonstrated forgetting/cost benefit

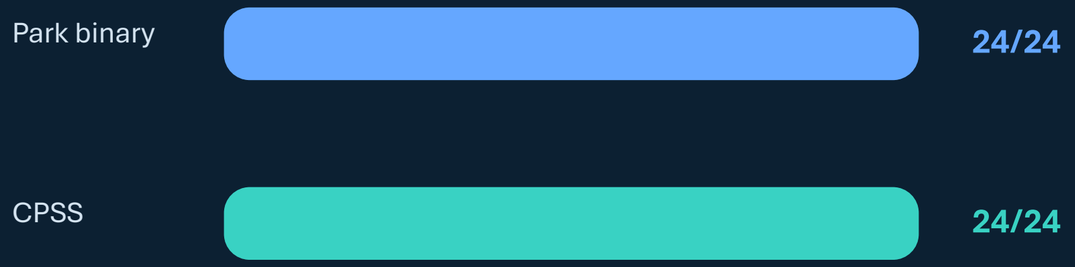
PILOT EVIDENCE

Park binary rule versus proposed CPSS

Archived exploratory synthetic AR(1) benchmark · 18 Oct 2026 · 24 held-out evaluation seeds per scenario.

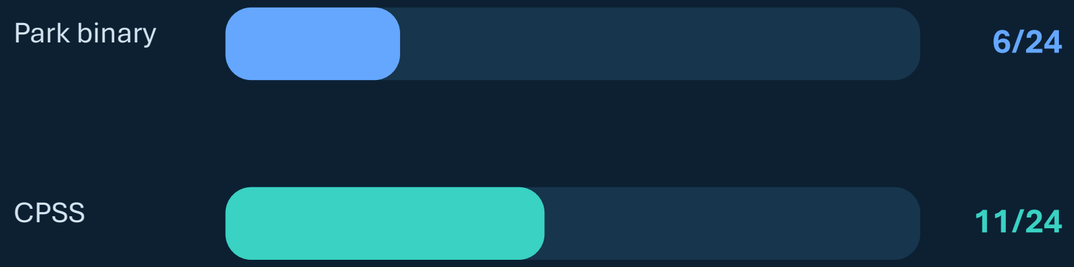
Standard mean shift 2.0

Mean-change detections out of 24



Weak mean shift 0.65

Mean-change detections out of 24



KEY LIMITATION CPSS stationary alarms 1/24 vs 0/24; 12/12 null calibration scores were zero. The weak-shift gain is not proof of improved discrimination at matched FPR.

Source: docs/park_cpss_benchmark.md · outputs/park_cpss/local_standard_01 & local_weak_01

NEXT FALSIFIABLE WORK

The research gaps point to testable experiments

The sequence respects chronological evidence, matched operating points and separate statistical targets.

01 Reconcile Raza’s EEG reproduction

Lock source/configuration/trial ordering; resolve B01 validation and 2A/2B counts.

EVENT TRACE · MANIFEST · EXACT COUNTS

02 Calibrate detection reliability

Independent null streams, alert burden, delay, Hotelling assumptions and matched false-alarm budgets.

FPR · DELAY · COVERAGE · INTERVALS

03 Test first- vs second-order changes

Separate mean, covariance and dependence regimes; compare Park, EWMA and high-dimensional alternatives.

POWER · CHANGE TYPE · LOCALISATION

04 Measure whether adaptation pays

Compare never, periodic, warning, validated, performance and evidence-value policies over time.

ACCURACY · RETRAINS · FORGETTING · COST

Status: proposed study design, not a report that these follow-up experiments have been executed.

SOURCE REGISTER

Representative publications & evidence locations

Clickable DOI / manuscript links; detailed annotations live in the Notion WP1 library.

Ramoser et al. (2000)	EEG / CSP
Raza, Prasad & Li (2015)	EWMA / CSD
Raza et al. (2016)	CSD adaptive BCI
Alippi et al. (2017)	Hierarchical CDT
Chowdhury et al. (2018)	Online neurorehab BCI
Raza et al. (2019)	CSE–UAEL
Gaur et al. (2019)	Tangent space / transfer

Diethe et al. (2019)	Self-maintaining ML
Wang et al. (2022)	L2P / DualPrompt
Kudithipudi et al. (2022)	Lifelong learning
Park et al. (2023)	Mean / second-order change
Hurtado et al. (2023)	CL / maintenance
Heyden et al. (2024)	ABCD / subspace
Gupta et al. (2024)	TWIN–ADAPT

Notion: Literature Topic Guide + 36-record library

GitHub: code, docs and archived output provenance

Park and Raza: two stages, different questions

Read Park et al. (2023), Section 2.1 and second-order methodology; compare Raza et al. (2015) and Chowdhury et al. (2018).

PARK et al. (2023)

Step 1: aggregated feature-wise mean-CUSUM.

- Estimate candidate split locations.
- If mean change is not established: investigate second-order structure using wavelets and dynamic PCA.

Step 2: second-order structural change detection

- Estimate wavelet-based, time-varying second-order structure.
- Apply DPCA to estimated structure to extract relevant dynamic components.
- Calculate CUSUM stats to estimate a structural change point.

RAZA / CHOWDHURY

Stage I: sequential EWMA score and control-limit warnings.

Stage II: statistical validation (Hotelling T^2 or two-sample K–S, depending on formulation).

Key boundary: retrospective CUSUM peak ≠ calibrated online alarm; EWMA warning ≠ confirmed shift.

PARK READING INSIGHT

Several CUSUM peaks ≠ several confirmed changes

Interpret peak structure as competing change-point evidence, not as a multimodal EEG probability distribution.

What the curves provide

$v_{\max(t)}$: strongest feature-wise normalised mean contrast at candidate t.

$v_{avg(t)}$: mean contrast across features.

Both identify candidate locations; their global maxima can disagree.

What needs investigation

Several local peaks could reflect heterogeneous feature responses, multiple changes, or stochastic fluctuations.

Inspect feature-wise traces, segmentation, dependence and metadata; do not infer an external cause from peaks alone.

A worked synthetic example illustrates candidate agreement	
Observed Result Planted change: 120 Smoothed peaks: 120 / 122 Estimated location: 121	**Evidence and limitations** Peak gap: 2, below the paper-rule threshold of 8.66 Both sampled unimodality counts: 0
Research question: Are estimated peak locations stable across repeated simulations and perturbations? Source: *Notebook 07, saved synthetic example; Park et al. (2023), Algorithm 1.*	

Sources: Park et al. (2023), DOI 10.29220/CSAM.2023.30.4.369; Fearnhead & Rigaill (2020), DOI 10.1002/sta4.291.

CODE-TO-MANUSCRIPT BOUNDARY

Park extension: implemented calculations versus open tests

Source: src/detection/stage1/park_mean_cusum.py and manuscript §§3.2, 4.8.

IMPLEMENTED

- Compute **feature-wise normalized CUSUM**
- Aggregate signals using **max and average statistics**
- Apply **local polynomial smoothing**
- Check for **unimodal peak structure**
- Verify **cross-method peak agreement**
- Calculate the **Change-Point Stability Score (CPSS)**

d: channel · T: observations · t: candidate change-point position

$$v_t^{(d)} := \left(\frac{1}{T}\sum_{u=0}^{T-1} X_{u,T}^{(d)}\right)^{-1} \left| \sqrt{\frac{T-t}{Tt}} \sum_{u=0}^{t-1} X_{u,T}^{(d)} - \sqrt{\frac{t}{T(T-t)}} \sum_{u=t}^{T-1} X_{u,T}^{(d)} \right|$$

Requires further evidence

- Robust false-alarm calibration
- Second-order wavelet/DPCA detection
- Real EEG change-point ground truth
- Matched-FPR comparisons
- Statistically validated CPSS advantage.

Research question RQ4: candidate localisation is implemented; confirmation and statistical validity remain open.

Park et al. (2023), Eq. 2.2

From Mean-CUSUM to Mean-Change Confirmation

PARK ET AL. (2023) / §2.3, ALGORITHM 1, STEPS 1–4

01 Aggregation Across Channels

$$v_t^{max} = \max_{1 \le d \le p} v_t^{(d)}$$

$$v_t^{avg} = \frac{1}{p} \sum v_t^d$$

Maximum and average combine channelwise mean-CUSUM evidence.

02 Smooth and Locate Peaks

$$\tilde{v}_t^g = LP_{0.1}(v^g)(t)$$

$$b_g = \arg\max_t \tilde{v}_t^g$$

Local polynomial span 0.1; peak locations from smoothed curves.

03 Check Unimodality

$$A^{max} A^{avg} = 0$$

k = 10 interior positions per side; at least one A count must be 0.

04 Confirm and Estimate

$$|b^{max} - b^{avg}| < \frac{\sqrt{T}}{2}$$

$$\hat{b} = \frac{b^{max} + b^{avg}}{2}$$

If both tests pass, estimate the mean change; else test second order.

Implemented in Python: Steps 1–4 | Statistical calibration & second-order detection remain open.

Source: Park et al. (2023), §2.3, Algorithm 1 | Code: src/detection/stage1/park_mean_cusum.py

Kronecker Delta: Index Equality and the Identity Matrix

Index equality gives a compact way to express identity and orthogonality.

$$\delta_{ij} = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

i and j label coordinates or components for comparison

The matrix $[\delta_{ij}]$ is the identity matrix. A delta expression alone does not specify a random variable's variance or independence.

$$I_3 = \left[\delta_{ij}\right]_{i,j=1}^{3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Source: Park et al. (2023) reading annotations, 12 October 2026; Kronecker-delta definition and standard linear-algebra interpretation.

Covariance and independence require separate assumptions

Zero-mean, unit-variance innovations can have identity covariance.

$$Cov(Z_i, Z_j) = E[Z_i Z_j] = \delta_{ij}$$

Diagonal entries equal 1 under unit variance.
Off-diagonal entries equal 0 under zero covariance.
Without zero means, $E[Z_i Z_j]$ also includes the product of the means.

Uncorrelated variables need not be independent.
For a jointly Gaussian vector, zero covariance between components implies independence.
The observed EEG channels can still be dependent.

Source: 12 October chat analysis. Illustrative standardised-innovation condition, not verified as Park’s exact notation.

A coordinate derivative describes sensitivity

The derivative in today’s notes and statistical variance answer different questions.

$$\frac{\partial w_i}{\partial w_j} = \delta_{ij}$$

The derivative is an entry of the identity Jacobian when w_i are independent coordinates.

$$Var(Z_i) = E[(Z_i - E[Z_i])^2]$$

Variance measures expected squared deviation from a random variable’s mean.

$$w_i(t+1) = w_i(t) + \eta e(t) x_i(t)$$

This is a separate online learning rule. Define the error convention and learning rate before interpreting it.

Source: user’s Park Notion notes, corrected through today’s mathematical discussion.

Daubechies wavelets resolve local variation

Wavelet scale and position describe where different patterns occur in time.

$$\int t^m \psi(t) dt = 0, \qquad m = 0, \ldots, N-1$$

ψ is the wavelet. N is its number of vanishing moments.
The integral vanishes for polynomial components below degree N, helping isolate local detail.
Compact support localises the wavelet in time.

Scale controls the duration of patterns.
Position controls their location.
Longer filters trade localisation for additional smoothness or vanishing moments.
Next: verify Park’s filter convention, scale range and boundary treatment.

Sources: Park Notion reading target; Wavelet background, Biometrika (2018), 105(4), 833–848. Specific Park settings remain to verify.

Transfer matrices can create observed dependence

Identity covariance of innovations does not force identity covariance of observations.

$$Y = AZ, \qquad Cov(Z) = I \quad \Rightarrow \quad Cov(Y) = AA^T$$

$$Y_1 = Z_1, \qquad Y_2 = aZ_1 + Z_2, \qquad Cov(Y_1, Y_2) = a$$

For a ≠ 0, the observed components have nonzero cross-covariance.
Time-varying transfer coefficients can change variance or dependence while leaving the mean unchanged. This motivates the second-order reading pathway.

Source: explanatory linear-transform identity. Park’s full locally stationary wavelet construction requires separate source verification.

PAPER CONTEXT

Chowdhury et al. (2018): online adaptive BCI

Calibration and task

10 healthy participants + 10 stroke patients
80 calibration trials, then 40 online trials
Left- versus right-hand motor imagery
Initial linear SVM and CSD parameters

Signal and adaptation

μ 8–12 Hz and β 16–24 Hz
CSP yields four log-variance features
PC1 EWMA warning, then Hotelling T^2
Validated shift triggers SVM retraining

Online output

Visual feedback and hand-exoskeleton movement follow MI classification.

Comparator: EEG-NAC replays the same recorded data offline with a fixed classifier.

Source: §II.B, p. 1072 · §II.D–E, pp. 1073–1075 · Figs. 1, 3 and 4

REPORTED ACHIEVEMENTS

Decoding improved relative to the fixed classifier

Group	EEG-NAC	EEG-CSAC	Gain
Healthy (n = 10)	75.25 ± 5.46%	81.50 ± 4.89%	+6.25 pp
Stroke (n = 10)	70.25 ± 3.43%	75.75 ± 3.92%	+5.50 pp

Both accuracy comparisons: $p < 0.01$

Kappa also improved significantly. Every participant could trigger the exoskeleton.

Source: §III, p. 1076, Tables II–IV and Fig. 6 · §IV, p. 1078

DETECTOR EVIDENCE

Stage-II validation reduced retraining triggers

Group	Stage-I CSW	Stage-II CSV
Healthy	15.75 ± 5.65%	9.25 ± 3.54%
Stroke	16.75 ± 2.89%	9.00 ± 2.10%

Rates are percentages of trials. Both reductions: $p < 0.01$.

Validated-shift rate and accuracy gain correlate: $r = 0.853$, $p < 0.01$.

Interpretation: fewer retraining triggers do not independently quantify false alarms.

Source: §II.E.4, p. 1075, Eqs. (10)–(11) · §III, p. 1077, Tables VI–VIII and Fig. 7

AUTHOR-STATED LIMITATIONS

Limitations acknowledged by the authors

Representation and comparison

Binary left/right MI classification
Fixed μ/β bands for all participants
No direct test against other adaptive BCIs

Labels and online control

Supervised adaptation assumes cue compliance
Synchronous BCI, one feedback event per trial
Continuous/asynchronous control remains future work

Artefacts and external noise

Online preprocessing adds latency. The authors acknowledge that contaminated EEG can reduce decoding accuracy.

Source: §IV, p. 1078 (scope, frequency bands, comparator, noise) · pp. 1078–1079 (control and labels)

DESIGN-DERIVED LIMITATIONS

Limits of the experimental evidence

Small cohort and short follow-up

10 participants in each group
One 40-trial online feedback run
Limited evidence across patient subgroups and repeated rehabilitation sessions

Comparison and recovery

Fixed baseline applied offline to the same data
No parallel online nonadaptive condition
Reported metrics concern BCI performance
Durable motor recovery remains untested

Supported conclusion

The study establishes online technical feasibility and improved decoding.
Longer-term reliability and rehabilitation benefit require further evaluation.

Source: §II.B, p. 1072 · §II.D–E, p. 1073 · §III, pp. 1075–1077

PAPER-TO-PROJECT MAPPING

Relevance to the current EWMA project

Methodological connection

EWMA warning followed by shift validation
Selective adaptation motivates event auditing
Paper uses two bands and four CSP features
Current deck uses 10-band FBCSP on 2A/2B

Existing next-step priorities

Lock the reproduction configuration
Explain accepted and rejected Stage-II events
Measure synthetic false alarms and delay
Export real EEG score/residual traces

Evidence boundary

Keep the paper’s online results separate from repository benchmarks. Numerical reproduction and real EEG extension evidence remain open in the current deck.

Sources: Chowdhury et al. (2018), §II.E and §III–IV; repository review at f3ef2e8.

FEATURE EXTRACTION

From EEG signals to the monitored PCA score

Two frequency bands: μ 8–12 Hz and β 16–24 Hz. Fit the transformations on calibration data.

01 Spatial filtering

$$Z = WE$$

Combine EEG channels using learned CSP filters.

E: one EEG trial; W: spatial filters; Z: filtered signals.

02 Log-variance features

$$f_j = log(\frac{Var(Z_j)}{\sum_{k=1}^{2m} Var(Z_k)})$$

Express each selected signal’s variance relative to the total.

The paper retains two features per band: four per trial.

03 PCA and PC1 selection

$$s_i = P^T(f_i - \mu_{cal}), \qquad X_i = s_{i,1}$$

Transform the feature vector, then monitor its first PC score.

The classifier uses features; Stage I monitors the scalar x_i.

Chowdhury et al. (2018), §II.E.1 and §II.E.3, Eqs. (1), (2), (4), pp. 1073–1074.
Indices clarified; PCA shown with explicit calibration centring. Code: fbcsp.py; detection/preprocessing.py.

REPRESENTATION VERSUS ADAPTATION

What is fixed and what changes during SVM updates

The stored training set grows and a new linear SVM is fitted; the feature and monitoring mappings are not refitted.

$$f_t = \phi_0(E_t), \qquad q_t = P_0^T(f_t - u_0), \qquad \hat{y}_t = g_{\theta_t}(f_t)$$

EEG → FBCSP

E_t is a trial; ϕ_0 is the mapping fitted at calibration. Not refitted during SVM updates.

FBCSP → PCA monitor

μ_0 and P_0 define the initial PCA coordinates. q_t is a monitoring score, not the SVM input.

FBCSP → prediction

$g\theta_t$ acts on f_t; θ_t changes when the SVM is retrained on accumulated examples.

Implementation boundary: refitting the classifier does not imply calling fit_fbcsp() or fit_cse_pca().

Fixed mapping means fixed fitted parameters — not constant features or scores over time.

How PCA scores become warnings and confirmations

Stage I: EWMA warning

State and one-step prediction error

$$z_i = \lambda x_i + (1-\lambda) z_{i-1}, \qquad e_i = x_i - z_{i-1}$$

Control limits use the previous variance

$$UCL_i = z_{i-1} + L\sigma_{e,i-1}, \qquad LCL_i = z_{i-1} - L\sigma_{e,i-1}$$

Smoothed error variance • Eq. (8)

$$\sigma_{e,i}^2 = \alpha_v {e_i}^2 + (1-\alpha_v)\sigma_{e,i-1}^2$$

Check current limits first; update state and variance afterwards.
α_v is the variance-smoothing weight (paper: 0.05).

Stage II: Hotelling validation

Repository pooled two-sample mode

$$T^2 = \frac{n_1 n_2}{n_1 + n_2} (\bar{x_1} - \bar{x_2})^T {S_p}^{-1} (\bar{x_1} - \bar{x_2})$$

Decision and dimensionality condition

$$p < \alpha_{test}, \qquad 2H - d - 1 > 0$$

H: observations per window; d: retained dimensions.
αtest is distinct from variance smoothing.
Confirmation is available after the second window.

Chowdhury §II.E.3, Eqs. (3), (5)–(9). Eq. (9) uses separate covariance terms; pooled mode is a repository choice.
Eq. (8) describes the always-update variance mode. The Diethe pilot uses frozen variance.

Two statistics called T^2 monitor different objects

A multivariate EWMA chart and a two-sample mean test use different reference structures.

$$T_t^2 = (z_t - \mu_0)^T \Sigma_{z,t}^{-1} (z_t - \mu_0)$$

Multivariate EWMA: compare a smoothed vector with a reference mean, using the covariance of that smoothed vector.

$$T^2 = \frac{n_1 n_2}{n_1 + n_2} (\bar{x_1} - \bar{x_2})^T S_p^{-1} (\bar{x_1} - \bar{x_2})$$

Pooled two-sample Hotelling: compare two window means. S_p is pooled within-window covariance. This mode assumes a common covariance and appropriate sampling conditions.

Source: standard statistical comparison, linked to the existing detector mathematics slide. Calibration assumptions must match the chosen method.

MATHEMATICAL READING UPDATE

Which structural changes do statistical tests target?

Conceptual comparison: mean, variance and temporal dependence require different statistical tests.

Detector design targets (conceptual) — evaluation outcomes NOT TESTED

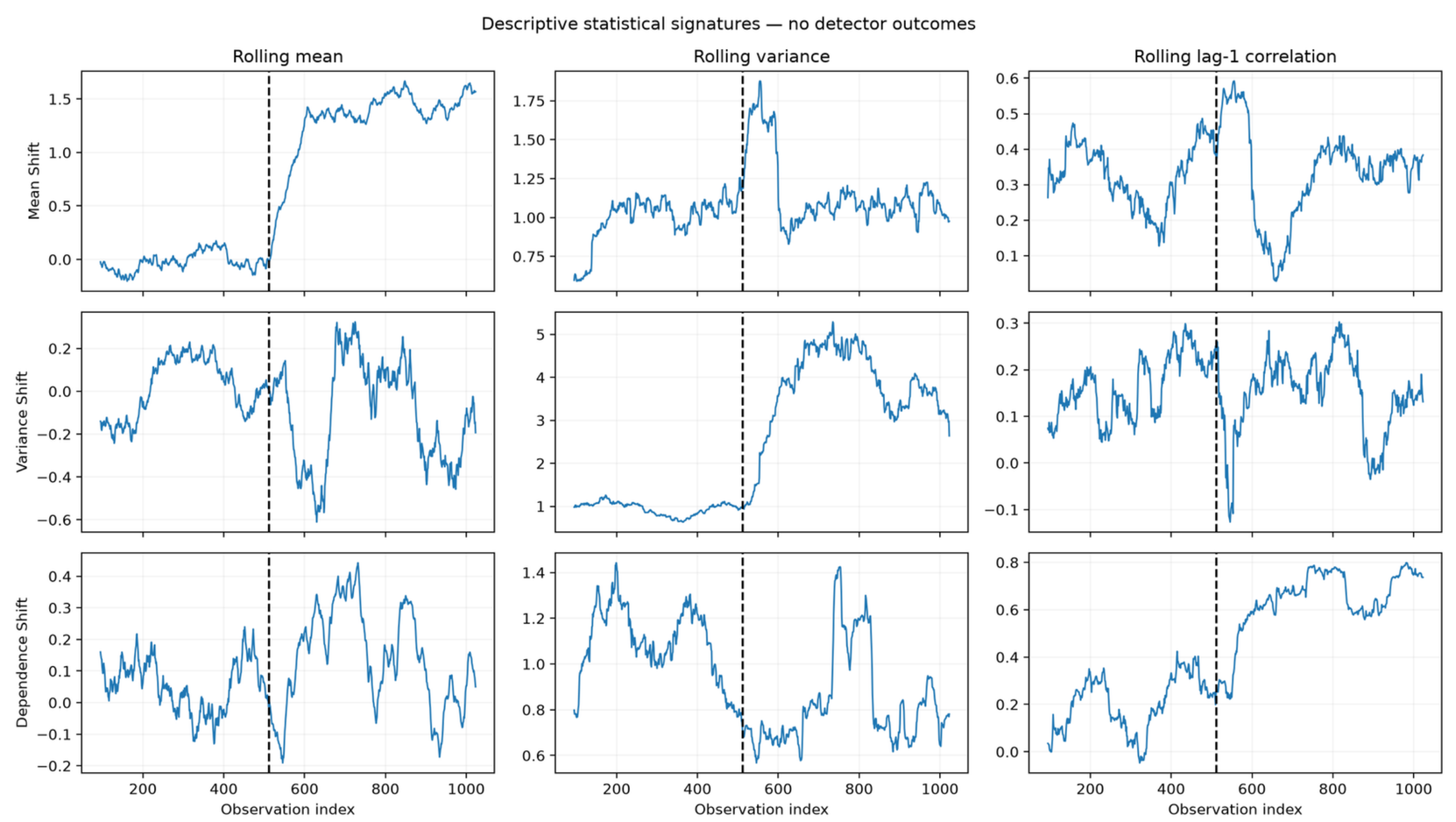
	Mean shift	Variance shift	AR dependence shift
Mean contrast / Hotelling T^2	Primary target	Not direct test	Not direct test
Variance comparison	Not direct test	Primary target	Not direct test
Lag/correlation test	Not direct test	Not direct test	Primary target

Conceptual test targets. Sources: Chowdhury et al. (2018), Eq. (9); Park et al. (2023), §3.1.

MATHEMATICAL READING UPDATE

How structural changes appear in rolling statistics

Mean, variance and AR dependence shift independently in controlled synthetic data.



ROWS
Mean shift
Variance shift
AR-dependence shift

COLUMNS
Rolling mean
Rolling variance
Lag-1 correlation

Dashed line: $t_o = 512$

KEY FINDING
Temporal dependence can shift without a mean shift. Mean-based validation cannot establish covariance stability.

Source: scripts/generate_slide22_structural_visuals.py · n = 1024 · $t_0 = 512$ · seed 42 · window 96. Synthetic, descriptive; detector outcomes untested.

Mean validation has a variance-change limitation

Covariance scaling adjusts the mean contrast but does not make it a direct covariance-equality test.
Different structural changes affect different statistical properties.

$$Q = (\bar{x_1} - \bar{x_2})^T (\frac{S_1}{n_1} + \frac{S_2}{n_2})^{-1} (\bar{x_1} - \bar{x_2})$$

Q and the pooled T^2 both start from a difference in means.
A pure variance or dependence change can therefore escape detection. Finite-sample mean fluctuations can still produce alarms.

Separate covariance terms need their own justified calibration. The pooled F conversion is not automatically exact.
Next comparison: mean-only, variance-only and dependence-only changes at matched false-alarm budgets.

Sources: existing detector slide; Chowdhury et al. (2018), Eq. (9). This expression preserves separate covariance terms.

How evidence becomes an update decision

Responsibility	Implemented behaviour
Representation	Cue-aligned EEG; FBCSP; PCA fitted on training data
Detection and validation	EWMA warnings; explicit Stage-II interpretations
Update policy	Never, periodic, warning, validated or performance-drop
Label regime	Supervised labels or confidence-gated PWKNN pseudo-labels
Audit outputs	Predictions, accepted/rejected evidence and update events

Score-plus-residual monitoring is a separate Stage-I extension.
The Diethe comparison currently uses the fixed CSE detector.

Repository evidence: 2a8d20b. Implemented capabilities and stored results are distinguished.

CSE TABLE-I DATA PROTOCOL

CSE development and held-out evaluation

BCI Competition IV 2A

Session I develops the model

Stratified 70/30 split within Session I
Fit FBCSP and PCA on training only
Transform validation with fitted models
Hold out Session II, left/right trials

BCI Competition IV 2B

Sessions I–III develop the model

Stratified 70/30 split within Sessions I–III
Fit FBCSP and PCA on training only
Transform validation with fitted models
Hold out Sessions IV–V

Shared rule

Evaluation trials do not enter feature-model fitting.
Record the split seed and effective detector configuration.

Classifier benchmark: 2A fits Session I and tests II.
The stored 2B benchmark fits I–II and tests labelled Session III.

Dataset roles differ across experiments

Experiment	2A	2B
Table-I CSE	Session I development; Session II evaluation	I–III development; IV–V evaluation
Stored classifier benchmark	Fit I; evaluate II	Fit I–II; evaluate III
Complementary EEG runner	I: training/calibration split; II evaluation	Fit I–II; calibrate III; evaluate IV–V
Diethe policy laboratory	Synthetic feature streams by default; prepared EEG features accepted by CLI	Raw EEG loading is not automatically integrated

Real EEG alarm traces describe alert burden. They do not supply ground-truth shift times.

Source: current runners and stored result files at 2a8d20b. Evaluation data do not fit the feature representation.

Diethe: separate change, performance and action

Question	Repository experiment
Has the input changed?	Shared CSE warnings and Stage-II validation records
Has prediction quality changed?	Rolling accuracy against a held-out validation reference
Should the model update?	Five explicit rules compared on the same stream
What does updating cost?	Update count, fitting time, prediction time and training size

Initial five-policy pilot is historical; a sixth evidence-sufficiency policy is now implemented.

Repeated matched comparisons across scenarios and seeds remain pending. This is not a Diethe reproduction.

Sources: Diethe et al. (2019), §§3–5; docs/diethe_experiments.md; src/adaptation/experiments/diethe.py.

SEPARATE EXPERIMENTAL CONTEXT

The Diethe policy lab uses supplied feature vectors

Do not conflate the real-EEG FBCSP/PCA monitoring pipeline with the synthetic policy-comparison branch.

Real-EEG branch

EEG trial → calibrated FBCSP → feature vector f_t
PCA is used by monitoring; SVM receives f_t.

Diethe laboratory branch

Supplied / synthetic features → classifier and CSE detector
No EEG extraction or FBCSP step is present.

Shared research question: should a confirmed distribution change trigger an update, and does that update help predictive performance?

Repository interpretation: run_policy_comparison(); the five-policy comparison is a controlled laboratory, not proof of clinical BCI benefit.

DIETHE EXPERIMENT DESIGN

Six policies, three controlled scenarios

Policy	Update trigger
Never / periodic	No update / every configured interval
Warning / validated	Stage-I warning / available Stage-II confirmation
Performance drop	Rolling accuracy drops below the validation reference

No shift: baseline alert and update burden.
Feature shift: P(X) changes; the label rule stays fixed.
Relationship shift: the label rule changes; P(X) stays fixed.

Predict before revealing each label. Training expands after updates.
Delayed labels, rollback and energy measurement remain outside this experiment.

Source: docs/diethe_experiments.md and current CLI. Defaults: 240 trials, 20-trial interval, 30-trial performance window.

ADAPTATION POLICY PROGRESS

Six policy families — one exploratory laboratory

Repository HEAD 98908f08f; Diethe-oriented comparison is not a reproduction of Diethe et al. (2019).

POLICIES

1. Never update (baseline)
2. Periodic retraining (passive)
3. EWMA warning-triggered update
4. Validated shift-triggered update
5. Performance-drop-triggered update
6. Evidence-sufficiency heuristic

VALIDATION BOUNDARIES

Three scenarios: stationary, marginal feature shift, changed label relationship.

Labels arrive immediately after prediction; historical training data accumulate.

Precision-gain proxy is not optimal welfare value. Repeated seeds, delayed labels, energy and forgetting remain open.

Sources: docs/diethe_experiments.md; src/adaptation/evidence_value.py; Diethe et al. (2019).

CONTINUAL LEARNING UPDATE

Hurtado-aligned passive updating: count new experience, not global trial position

Source: Hurtado et al. (2023), §4.1.2; policy refinement for src/adaptation/policies.py

BEFORE: CLOCKED SCHEDULE

Absolute trial index

```
(trial_index + 1) % interval == 0
```

Can retrain soon after a separate update event.

POLICY UPDATE



semantics

AFTER: PASSIVE STREAM BASELINE

New data since last update

```
trials_since_update >= interval
```

Updates only after enough fresh streaming observations.

Why this matters for PdM

Hurtado’s passive side assumes a continuously evolving stream.

Cleaner experiment axis

Separate when to update from which data or labels enter training.

Next passive baseline

Add ContinuousRetrain(batch_size) for mini-batch or per-observation updates.

Achievement: policy taxonomy now distinguishes active triggers from passive stream updates, supporting continual-learning experiments for predictive maintenance.

hlin863/ewma_covariate_shift_demo

A detected shift does not guarantee a useful update

Synthetic feature shift; seed 42; 240 trials; shift starts at index 120.

Policy	Accuracy	Updates	Gain vs never
Never	97.92%	0	0.00 pp
Periodic	98.33%	12	+0.42 pp
Warning	97.50%	3	−0.42 pp
Validated	97.50%	1	−0.42 pp
Performance drop	97.92%	0	0.00 pp

Periodic updating gains one correct prediction overall for 12 updates.
Validated updating finishes one prediction below never-update.

One-run illustration, not a policy ranking. Next: matched comparisons across scenarios and seeds.

Source: outputs/diethe/20261004T111759599003Z/summary.csv; generating commit 2a8d20b.

TRACKED EXPERIMENT RESULTS

Classifier and complementary-monitoring results

Real EEG classifier benchmark

Mean accuracy across 9 subjects per dataset

Dataset	Single SVM	Bagged SVM
2A	76.00%	75.85%
2B	74.86%	75.42%

Mean change: −0.15 pp (2A), +0.56 pp (2B).
No consistent gain across both datasets.

2A: Session II evaluation
2B: Session III GDF-only holdout

Complementary Stage-I monitoring

Detection within 20 samples, 100 seeds, magnitude 3

Shift	Score	Residual	Combined
PC1	35%	21%	31%
PC2	21%	100%	100%
Both	26%	100%	99%

Magnitude 6: PC1 detection rises to 88% (score) and 82% (combined).

No-shift 20-sample alarm rate: 21–26%.
Per-sample false-alarm rate: about 1%.

$$Q_t = dL_t$$

CI at 2a8d20b: 255 passed, 1 failed. Homepage test expects /diethe; route is /results/diethe.

B01: two saved runs give different validation totals

B01 evidence	Warnings	Validated shifts
Published reference	14	10
Standalone diagnostic	17	5
Combined Table-I file	17	0

What this means

Both stored runs differ from the published result. Neither file proves that one run supersedes the other.

Next: record the Stage-II mode, window settings and generating commit for each run.

Sources: outputs/metrics/bci_2b_table1_results.csv and bci_table1_comparison.csv at 2a8d20b.

Numerical reproduction remains unresolved

Mean validated-shift count	Computed	Published
Dataset 2A	0.33	5.56
Dataset 2B	0.67	10.33

Evidence available

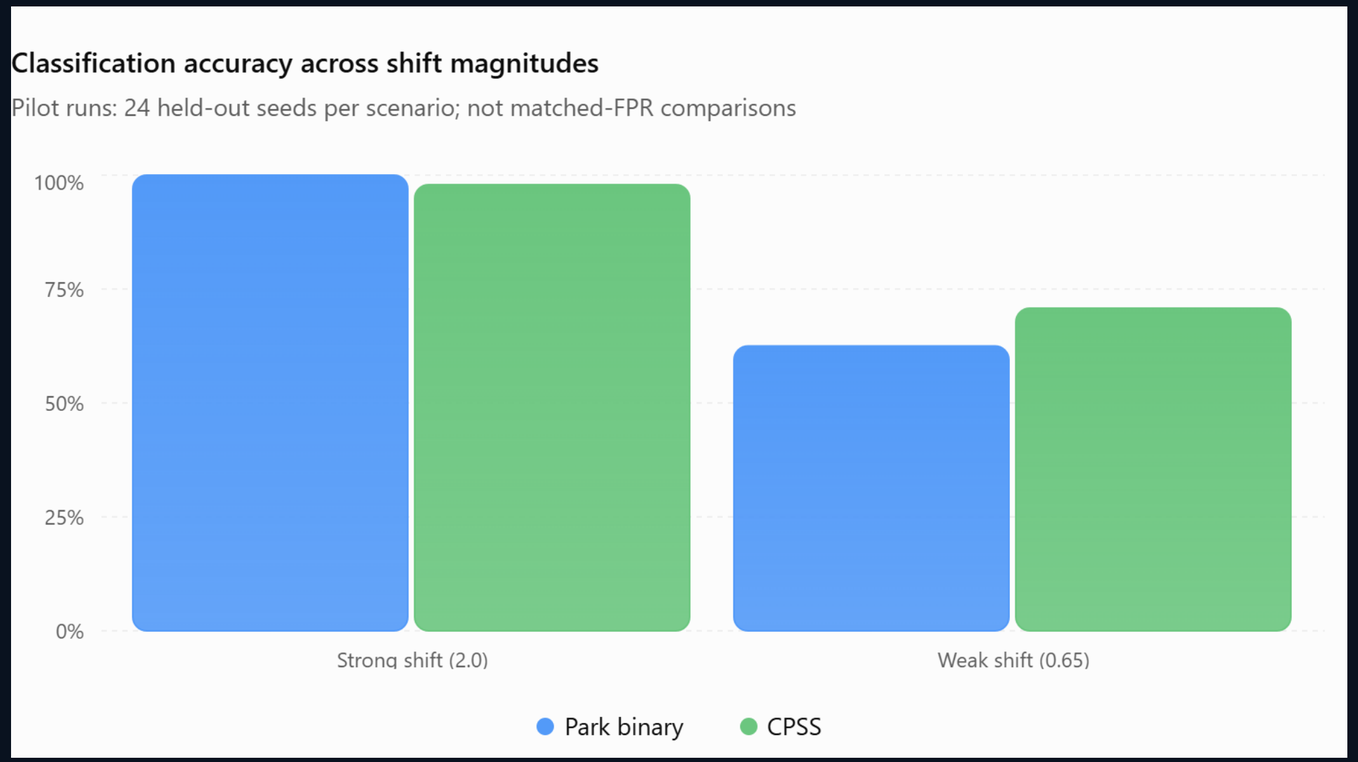
Subject-level comparisons
Classifier benchmarks
Synthetic score/residual ablations

Evidence to complete

One configuration per reproducible run
Stage-II acceptance/rejection analysis
Real EEG traces and repeated Diethe runs

Source: outputs/metrics/bci_table1_comparison.csv at 2a8d20b; means calculated across nine subjects per dataset.

CPSS versus Park Binary



Weak-shift arithmetic	Park binary	CPSS
Accuracy	62.5 %	70.8 %
Sensitivity	25.0 %	45.8 %
F1	0.400	0.611
ROC AUC	0.625	0.709
Stationary false alarms	0/24	1/24

Preliminary finding

CPSS detects 11/24 weak shifts versus 6/24 for Park but introduces one stationary false alarm.

Limitation: The null calibration produced $\tau = 0$. The apparent improvement may reflect a more permissive detection threshold rather than statistically validated superiority.

hlin863/ewma_covariate_shift_demo

CSE reconstruction and policy prototypes implemented

Workstream	Current position
CSE reconstruction	EEG features, EWMA warnings and Stage-II diagnostics
Supervised adaptation	Predict first, then update using the revealed label
Transductive adaptation	PWKNN pseudo-labels; confidence-gated training updates
Policy experiments	Five Diethe-inspired policies on matched streams
Evidence still needed	Matched reproduction runs and saved adaptation comparisons

Full dynamic CSE-UAEL ensemble growth and weighted voting remain open.

PROJECT TIMELINE

Year 1 timeline · Part 1: WP1–WP3

October 2026 – September 2027 | Foundations, detection evidence and adaptation

Oct–Dec 2026 · **Jan–Mar 2027** · **Apr–Jun 2027** · **Jul–Sep 2027**

planning horizon; exact package dates not yet validated

WP1 · Theory and architecture

CURRENT EVIDENCE

EWMA / Stage-II pipeline, research architecture drafted

NEXT DELIVERABLE

Documented CPD–CL–digital-twin architecture

WP2 · CPD benchmarking

CURRENT EVIDENCE

Mean-CUSUM smoothing, Park simulation models and controlled scenario visualisations

NEXT DELIVERABLE

Calibrated multi-seed benchmarks and Stage-II gap investigation

WP3 · Continual learning and sustainability

CURRENT EVIDENCE

Six adaptation-policy prototypes and supervised/pseudo-label experiments

NEXT DELIVERABLE

Forgetting, replay, update-cost and energy evaluation

Evidence boundary: Implemented code # validated detection or reproduction; BOCPD, density ratios and energy benchmarking remain planned.

Year 1 timeline · Part 2: WP4–WP6

October 2026–September 2027 · planned work packages.

Foundation → **Evaluation** → **Synthesis**

WP4 · Digital twin co-evolution

Joint updates, disagreement, safety

Deliverable: twin framework and testbed

WP5 · Cross-domain evaluation

BCI and phonic maintenance

Deliverable: comparative transfer benchmarks

WP6 · Synthesis and engagement

Manuscript, open research artifacts, stakeholder review

Deliverable: submission and release roadmap

Dependency: verified CPD and repeated adaptation benchmarks precede twin safety evaluation and cross-domain claims.

Next milestone: a reproducible evidence package

1. Reconcile the saved reproduction runs

Lock settings and explain the Stage-II accepted, rejected and ineligible events.

2. Generate adaptation and real EEG evidence

Extend the saved Diethe pilot across three scenarios and 20 seeds. Export 2A/2B complementary traces.

3. Align tests, slides and research notes

Resolve the homepage route assertion. Refresh Notion implementation status and link every numerical claim to its run.

Repository evidence: 2a8d20b. Implemented capabilities and stored results are distinguished.

Reading, writing and project management reinforce each other

Reading

Strength: interrogating symbols and connecting methods.

Next: record each symbol’s role, assumptions and equation location.

Writing

Strength: maintaining explicit evidence boundaries.

Next: separate mathematical identity, modelling assumption and empirical finding.

Project management

Strength: linking notes with code and a research diary.

Next: keep one dated claim ledger and define a falsifiable test for each open question.

Sources: supplied learning image and today’s chat. Assessment reflects discussion evidence, not a scored evaluation.